

Local to Azure Databricks Mapping

Local Stack to Azure Databricks Enterprise Mapping

This document maps every tool used in the NorthPeak simulation to its enterprise equivalent on Azure/Databricks. Use this in interviews to demonstrate you understand the production context behind the local implementation.

Compute and Processing

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

| PySpark 3.5 local mode | Databricks Runtime 14.x on Azure | Same API. Databricks adds Photon engine, auto-scaling clusters, GPU support |

| `spark.read.parquet()` | Same in Databricks | Identical code |

| Local `SparkSession` | Databricks cluster `SparkSession` | Cluster managed by Databricks - no setup needed |

Storage and Table Format

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

| Delta Lake OSS (delta-rs 0.18) | Databricks Delta Lake | Databricks adds OPTIMIZE ZORDER, Vacuum automation, Delta Sharing |

| Local Parquet files | Azure Data Lake Storage Gen2 (ADLS) | ADLS adds access control via Azure Active Directory, encryption at rest |

| DuckDB 0.10 (warehouse) | Snowflake / Azure Synapse Analytics | Scale: DuckDB handles GB, Synapse handles PB |

Orchestration

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

| Apache Airflow 2.9 (Docker Compose) | Databricks Workflows / AWS MWAA / Google Cloud Composer | Managed Airflow removes infrastructure burden |

| Local DAG files | Same DAG files deployed to managed Airflow | DAG code is identical |

| Airflow retry config | Same retry/alert config in Databricks Workflows | Databricks Workflows uses JSON task config instead of Python decorator |

Transformation

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

dbt Core 1.8 (free)	dbt Cloud / Databricks dbt integration	dbt Cloud adds IDE, scheduler, CI/CD, metadata sync
dbt + DuckDB adapter	dbt + Snowflake / Databricks / Synapse adapter	Change one line in profiles.yml
Local dbt docs	dbt Cloud docs hosting	Same HTML output, hosted automatically

Data Quality

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

Great Expectations 0.18	Monte Carlo / Acceldata / GE Cloud	Enterprise tools add ML-based anomaly detection, Slack/PagerDuty integration
YAML expectation suites	Same concept in enterprise tools	GE OSS suites are portable to GE Cloud
pytest pipeline tests	Same pytest in CI/CD	No difference in test code

CI/CD and Infrastructure

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

GitHub Actions (free tier)	Azure DevOps Pipelines / Jenkins	Same YAML structure. Azure DevOps adds service connections, environments
Terraform CLI plan output	Azure provider full deployment	Same HCL syntax. Production adds state backend (Azure Storage), remote planning
Local .env secrets	Azure Key Vault / AWS Secrets Manager	Never commit secrets. Enterprise uses managed identity

Governance

| Simulation | Enterprise | Key Difference |

|-----|-----|-----|

YAML data contracts	Unity Catalog (Databricks) / Collibra / Alation	Unity Catalog enforces contracts at query level
Custom logging	Azure Monitor / Datadog / CloudWatch	Enterprise tools add dashboards, alerting, cost attribution
Manual access control docs	Azure RBAC + Unity Catalog permissions	Programmatic enforcement at storage and compute level

Interview Talking Points

"We used PySpark locally, but the same code runs unchanged on Databricks. I designed the pipeline to be cloud-portable from day one by avoiding local file paths and using Delta format which Databricks natively supports."

"Our Airflow DAGs would deploy directly to AWS MWAA or Cloud Composer because we followed standard operator patterns and kept all connection strings in environment variables, not hardcoded."

"The dbt project targets DuckDB locally but changing one profile setting migrates it to Snowflake or Databricks SQL. The model logic is warehouse-agnostic."